

UNIVERSIDADE ESTADUAL DE MONTES CLAROS – UNIMONTES
CENTRO DE CIÊNCIAS EXATAS E TECNOLÓGICAS – CCET
DEPARTAMENTO DE CIÊNCIAS DA COMPUTAÇÃO – DCC

ATIVIDADE XXI

RAMON LOPES DE QUEIROZ

MONTES CLAROS – MG
JUNHO/2026

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Atividade avaliativa apresentada para atendimento de requisito parcial para aprovação na disciplina Cálculo Integral e Diferencial do Curso de Graduação em Bacharelado em Sistemas de Informação – 2º período

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Atividade 21.

1-a. $\int (3x-2)^3 dx$ $\int u^3 \frac{du}{3} = \frac{1}{3} \int u^3 du \Rightarrow \frac{1}{3} \frac{u^4}{4} = \frac{u^4}{12} = \frac{(3x-2)^4}{12} + C$
 $u = 3x-2$
 $du = 3dx \rightarrow dx = du/3$

b. $\int \sqrt{3x-2} dx$ $\int u^{1/2} \cdot \frac{du}{3} = \frac{1}{3} \int u^{1/2} du = \frac{1}{3} \cdot \frac{2}{3} u^{3/2} = \frac{2}{9} \sqrt{(3x-2)^3} + C$
 $u = 3x-2 \rightarrow du = 3dx$

c. $\int \frac{1}{3x-2} dx = \int \frac{1}{u} \cdot \frac{du}{3} = \frac{1}{3} \int \frac{1}{u} du = \frac{1}{3} \ln|u| = \frac{1}{3} \ln|3x-2| + C$
 $u = 3x-2$
 $dx = du/3$

d. $\int \frac{1}{(3x-2)^2} dx = \int u^{-2} \cdot \frac{du}{3} = \frac{1}{3} \int u^{-2} du = \frac{1}{3} \left(\frac{u^{-1}}{-1} \right) = -\frac{1}{3u} = -\frac{1}{3(3x-2)} + C$
 $u = 3x-2$
 $dx = du/3$

e. $\int x \cdot \sin(x^2) dx$ $\int \sin u \cdot \frac{du}{2} = \frac{1}{2} \int \sin u du = \frac{1}{2} (-\cos u) = -\frac{1}{2} \cos x^2 + C$
 $u = x^2$
 $du = 2x dx \rightarrow x dx = du/2$

f. $\int x \cdot e^{x^2} dx = \int e^u \cdot \frac{du}{2} = \frac{1}{2} \int e^u du = \frac{1}{2} e^u = \frac{1}{2} e^{x^2} + C$
 $u = x^2$
 $x dx = du/2$

$$g. \int x^3 \cdot e^{x^3} dx$$

$$u = x^3$$

$$du = 3x^2 dx \Rightarrow dx = \frac{du}{3}$$

$$\int e^u \cdot \frac{du}{3} = \frac{1}{3} \int e^u du = \frac{1}{3} e^u = \frac{1}{3} e^{x^3} + C$$

$$2.1. a. \int x e^x dx = x \cdot e^x - \int e^x dx \Rightarrow x e^x - e^x + C$$

$$u = x$$

$$du = dx$$

$$dx = e^x dx \Rightarrow \int e^x dx = e^x$$

$$b. \int x \sin x dx = x \cdot (-\cos x) - \int (-\cos x) dx$$

$$u = x \rightarrow du = dx$$

$$= -x \cos x + \int \cos x dx$$

$$dx = \sin x dx \rightarrow \int \sin x dx = -\cos x$$

$$= -x \cos x + \sin x + C$$

$$c. \int x^2 e^x dx = x^2 e^x - \int e^x (2x dx)$$

$$u = x^2 \rightarrow du = 2x dx$$

$$= x^2 e^x - 2 \int x e^x dx$$

$$dx = e^x dx \rightarrow \int e^x dx = e^x$$

$$= x^2 e^x - 2(x e^x - e^x)$$

$$= x^2 e^x - 2x e^x + 2e^x + C$$

$$d. \int x \ln x dx = \int x \ln x dx = (\ln x) \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx$$

$$u = \ln x \quad dv = x dx$$

$$du = \frac{1}{x} dx$$

$$v = \frac{x^2}{2}$$

$$= \frac{x^2}{2} \ln x - \frac{1}{2} \int x dx$$

$$= \frac{x^2}{2} \ln x - \frac{1}{2} \left(\frac{x^2}{2} \right)$$

$$= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$$

3
a. $\int \sqrt{1-4x^2} dx = \int \cos \theta \cdot \left(\frac{1}{2} \cos \theta d\theta \right) = \frac{1}{2} \int \cos^2 \theta d\theta$
 substitution: $2x = \sin \theta$

$$dx = \frac{1}{2} \cos \theta d\theta$$

$$\sqrt{1-\sin^2 \theta} = \sqrt{\cos^2 \theta} = \cos \theta$$

$$= \frac{1}{2} \int \frac{1 + \cos(2\theta)}{2} d\theta = \frac{1}{4} \left(\theta + \frac{\sin 2\theta}{2} \right) = \frac{1}{4} \theta + \frac{1}{8} \sin 2\theta$$

$$2x = \sin \theta \rightarrow \theta = \arcsin 2x$$

$$\frac{1}{4} \arcsin 2x + \frac{1}{8} x \sqrt{1-4x^2} + C$$

b. $\int \frac{1}{\sqrt{4-x^2}} dx = \int \frac{2 \cos \theta}{2 \cos \theta} d\theta = \int 1 d\theta = \theta = \arcsin \left(\frac{x}{2} \right) + C$

$$x = 2 \sin \theta$$

$$\sin \theta = x/2$$

$$dx = 2 \cos \theta d\theta$$

$$\sqrt{4-4\sin^2 \theta} = 2 \cos \theta$$

c. $\int \frac{1}{4+x^2} d\theta = \int \frac{2 \sec^2 \theta}{4 \sec^2 \theta} d\theta = \frac{2}{4} \int 1 d\theta = \frac{1}{2} \theta = \frac{1}{2} \arctan \left(\frac{x}{2} \right) + C$

$$x = 2 \tan \theta \rightarrow dx = 2 \sec^2 \theta d\theta \quad \tan \theta = x/2 \quad \theta = \arctan(x/2)$$

$$4 + 4 \tan^2 \theta = 4 \sec^2 \theta$$

Ex 8. $A = 4 \int_0^n \sqrt{n^2 - x^2} dx \Rightarrow n \int_0^n \sqrt{1 - \left(\frac{x}{n}\right)^2} dx \Rightarrow n \int_0^{\pi/2} \sqrt{1 - \sin^2 u} \cdot n \cos u du \Rightarrow$

$$x = n \sin u \rightarrow dx = n \cos u du$$

$$x=0 \rightarrow u=0$$

$$x=n \rightarrow u = \frac{\pi}{2}$$

$$n^2 \int_0^{\pi/2} \cos^2 u du \Rightarrow \int_0^{\pi/2} \left(\frac{1}{2} + \frac{1}{2} \cos 2u \right) du \Rightarrow$$

$$n^2 \left[\frac{1}{2} u + \frac{1}{4} \sin 2u \right]_0^{\pi/2} = \frac{\pi n^2}{4}$$

$$A = 4 \int_0^n \sqrt{n^2 - x^2} dx = \pi n^2$$